

Political Bias

Coreference | NER | Stance

NLP PIPELINE

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Problem Overview

STANCE VS SENTIMENT

"The mayor's opponent is an absolute disaster"

The goal of this project is to build an NLP pipeline that **detects**, **quantized** and **visualizes** political bias in news media by analyzing how different articles portray the same political entities.

Standard sentiment analysis often fails on news text because it scores the overall mood of a sentence rather than the author's stance toward a specific subject. By combining Named Entity Recognition with Aspect-Based Sentiment Analysis, this project aims to narrow political figures and measures the targeted **STANCE** directed at them, then to computationally identify media bias.

Motivation:

Using News articles

Author's **Stance** of every subject

Visualize Author bias

Goal:

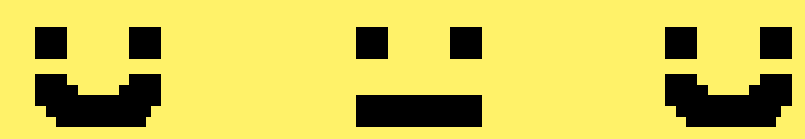
Get people as **Subject**

Coreference

Determine **Stance**

Visualize Data

Motivation & Goals



Dataset Description

AllSides

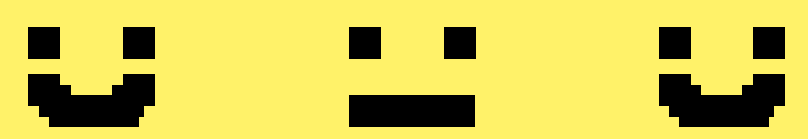
Used to reference for publisher orientation. The dataset provides bias ratings more than 1,400 news outlets, allowing me to select "opposing" articles for comparison.

BASIL

A specialized dataset of 300 news articles organized into "triplets". More than 1,700 manual annotations for lexical and informational bias.

WikiData

a free, collaborative, and multilingual, structured knowledge base that acts as central storage for structured data for projects and other external sites



Approach Overview

Get SUBJECT

COREFERENCE

STANCE

SpaCy NER - Entity Linking

Model : en_core_web_trf

resolves ambiguous textual mentions
Entity Link using Wikidata
unique identifiers in a Knowledge Base
Filter for **PERSON** only
Add to Subjects_ Dictionary

FCoref

Model : fastcoref

Link pronouns (ei, "he," "the President")
back to their root entities
Filter **Subjects** in entities
Replace text span with linked Subject

Hugging Face

MoritzLaurer/DeBERTa-v3-large-zeroshot-v2.0

NLI - Natural Language Interface
Hypotheses :
"What is the author's stance to
{Subjects.Name} In sentence"
NEGATIVE , **POSITIVE** , **NEUTRAL**

Tools & Techniques

Before any stance can be identified, the text must undergo thorough preprocessing and normalization.

AI & Core NLP - Python
spaCy: Named Entity Recognition (NER).
fastcoref (FCoref)
DeBERTa-v3-large-zeroshot-v2.0

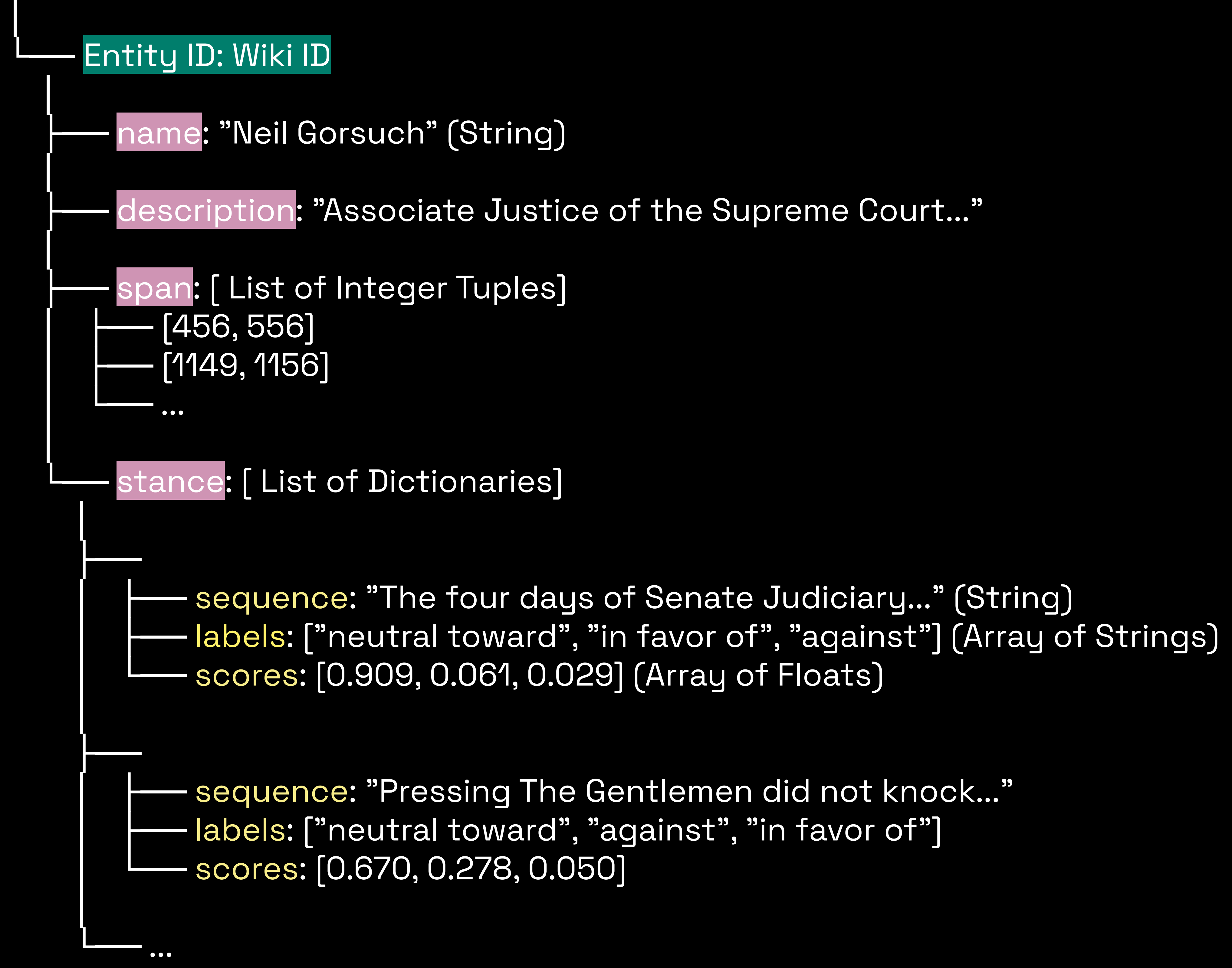
Backend API & Data Processing
Pandas
Flask
JSON

Frontend UI & Visualization (TypeScript)
React
Chart.js

Issues	Solutions
Lost Context (The Coreference Problem) Ignore subsequent pronouns	Context Preservation via Coreference Fastcoref
Alias & Disambiguation Collisions multiple aliases for one target Subject "Donald Trump", "DJT" "George Bush", "Jeb Bush"	Entity Linking via spacy-entity-linker
String Matching Failures Map coreference clusters to entities	Mapping via Character Index Overlap
Grammar vs. Contextual Meaning	
The Dilution Effect flat average of stance	Ruling from meaning and context

Experiments & Analysis

Subject Dictionary



Original Sentence

Sen. Sheldon Whitehouse (D-R.I.) made this point abundantly clear when he pointed to the series of 5-4 decisions made under Roberts 2019 leadership that have changed laws to help Republicans get elected, big donors gain more influence in politics through campaign contributions and corporations gain more economic and political power.

Conferenced and Linked Sentence

Subject: John Roberts **Label:** Against **Score** 0.8758
Sheldon Whitehouse made this point abundantly clear when Sheldon Whitehouse pointed to the series of 5-4 decisions made under John Roberts leadership that have changed laws to help Republicans get elected, big donors gain more influence in politics through campaign contributions and corporations gain more economic and political power.

Original Sentence

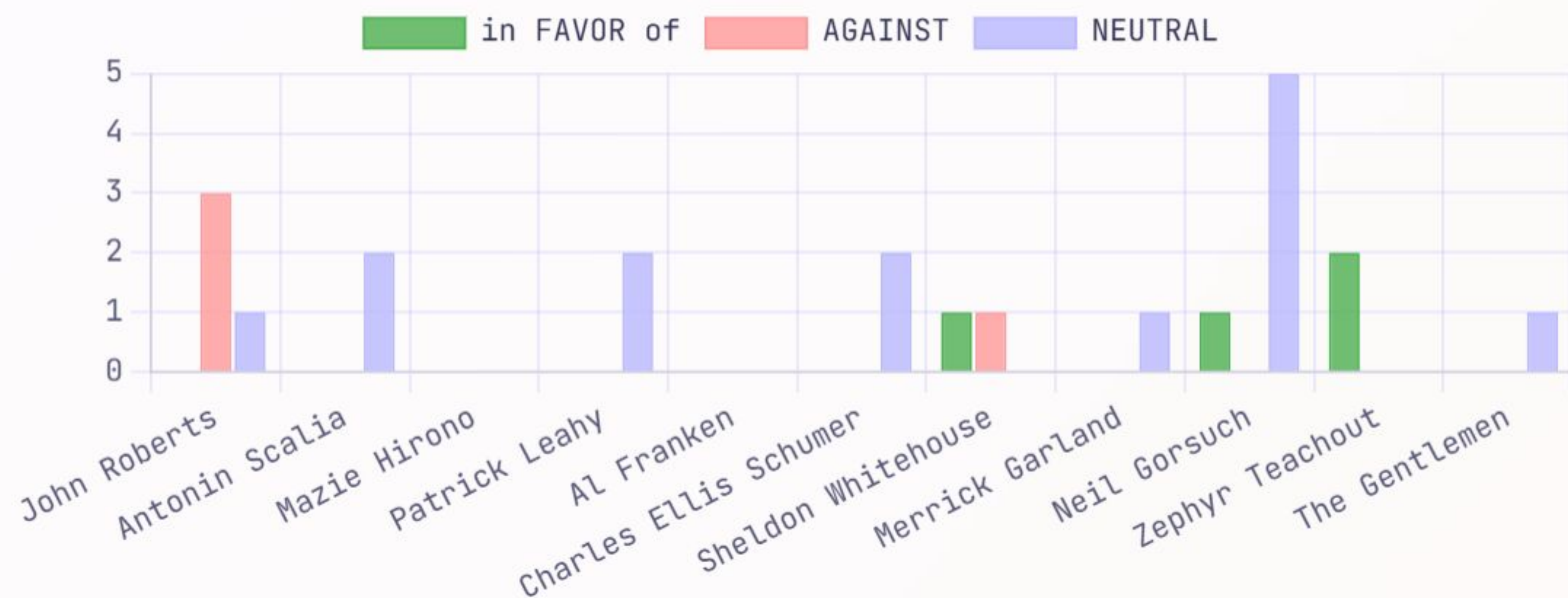
He understands that if you ease off before the dynamic shifts, you don't get a better deal — you get played.

Conferenced and Linked Sentence

Subject: Donald Trump **Label:** FAVOR **Score** 0.869
Donald Trump understands that if you ease off before the dynamic shifts, you don't get a better deal — you get played."

Results

Political Stance by Person



Sentiment Data:

FAVOR

AGAINST

NEUTRAL

Confidence Score: 0.7

Subjects:

John Roberts

Antonin Scalia

Mazie Hirono

Patrick Leahy

Al Franken

Charles Ellis Schumer

Sheldon Whitehouse

Merrick Garland

Neil Gorsuch

Zephyr Teachout

The Gentlemen

Detailed Subject Review

Select Subject: Mitch Landrieu

Sentence Confidence Score: 0.7

Name: Mitch Landrieu

Mentions (Span count): 82

Description: American politician

Stances (35)

IN FAVOR OF Score: 0.864

"It helped that the meeting as in the city that gave Mitch Landrieu the accent with which Mitch Landrieu squeezes three soft syllables out of the second word in "New Orleans.""

NEUTRAL TOWARD Score: 0.731

"It's like when New Orleans got destroyed by Katrina," Mitch Landrieu told a gathering of the Young Democrats of America attached to an otherwise ho-hum seasonal meeting of the Democratic National Committee this month."

NEUTRAL TOWARD Score: 0.893

"Mitch Landrieu was riffing without notes."

Challenges & Lessons Learned

Text Normalization

Grammar, Case sensitivity, Slang

Granularity of Context

Document-level coreference
vs strictly Sentence-level
Stance

Subject Text Replacement

Span vs Fuzzy

The Entity Collision

“Donald Trump’s nominee to fill the late Justice Antonin Scalia’s seat, Judge Neil Gorsuch”

Entity Hallucinations

Categorization Failures

Clean Data

BASIL vs. The Wild

Data Visibility

User Adjust Data

API

Time
constraint

Conclusion

Measuring true political bias requires a multi-model pipeline. Human language relies so heavily on assumed context and relational grammar that no single model can capture it alone.

Future Work

1. Add NORG and GPD

Next step is expanding the entity extraction to track bias against ORG (Democrats, Republicans) or Geolocations (New York City, Minnesota)

2. Live Data Integration

hooking the backend up to a live news API (NewsAPI) to create an automated daily tracker of media bias across different sources.